

Nitrogen deposition controls intrinsic water-use efficiency in *Fagus sylvatica* regeneration more tightly than climate

Markus Hauck^{1,*}, Viktoria Dietrich²

Applied Vegetation Ecology, University of Freiburg, Stefan-Meier-Str. 76, 79104 Freiburg, Germany

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ABSTRACT

Intrinsic water-use efficiency (WUE_i) is an important parameter quantifying the tree's ability to maintain the balance between transpiration and photosynthesis. It can be derived from foliar $\delta^{13}\text{C}$ signatures and depends on the carboxylation efficiency and stomatal conductance (g_s). Because of the control of the temporal variation of g_s by soil moisture and the atmospheric vapor pressure deficit (VPD), the interpretation of WUE_i and g_s in the ecological literature is often confined to tree water relations, and tree water relations are receiving increased attention in the light of climate change. However, it is also known that WUE_i and g_s can be additionally influenced by other factors, including N. Here we show in a field survey with European beech (*Fagus sylvatica*) regeneration covering gradients of mean annual precipitation (MAP) of 1130 mm (560 – 1690 mm), of mean annual temperature (MAT) of 3.4 °C (6.8 – 10.2 °C), and of atmospheric N deposition of 21 kg N ha⁻¹ a⁻¹ (10.5 – 31.4 kg N ha⁻¹ a⁻¹) that the variability of WUE_i and foliar $\delta^{13}\text{C}$ signatures across 45 forest stands that were distributed over most of Germany was more closely related to N deposition than to MAP or MAT. The results suggest that stomates were kept more open at high N deposition. This result underlines the importance of interactions between plant N and water relations and points to anthropogenic N deposition as an additional risk factor for trees that may exacerbate drought and heat stress due to climate change.

1. Introduction

Vitality and productivity reductions in trees are widespread consequences of climate change in all forested biomes that are not outweighed by growth releases and forest expansions, which partially occur under cold and well-water supplied climates (Allen et al., 2010; Hauck et al., 2019). Research on the causalities of these changes in forest health and productivity has been strongly focused on the drought vulnerability of trees (Mantova et al., 2022). Only recently it has been recognized that the heat tolerance of trees deserves greater attention as an independent factor of climate change effects in addition to drought tolerance (Kunert et al., 2022; Hauck et al., 2025). Biotic interactions, where drought and heat promote herbivores and pathogens, are also significant and well studied (Guégan et al., 2023; Dobor et al., 2024).

Interactions between nutrient availability and drought effects are much less discussed, even though the existence of such interrelations has long since been documented (Kreuzwieser and Gessler, 2010; Sardans et al., 2017). Especially N is tightly interrelated with plant water

relations. The N availability affects biomass allocation to the water-transpiring foliage or the roots that function for water and nutrient uptake (Schulze et al., 1994; Li et al., 2020; Dietrich et al., 2025). Furthermore, N directly interferes with plant water relations by controlling stomatal conductance (g_s), because not only low soil moisture and high atmospheric vapor pressure deficits (VPD), but also low N availability can induce abscisic acid (ABA) production and thus reduce g_s (Chapin et al., 1988; Mu et al., 2018). The evolutionary background of this relationship between N and g_s is probably to be seen in a slowing-down of growth in the case of nutrient deficiency in order to counteract deficiency symptoms (Chapin, 1991). What made sense in a world limited by N deficiency can, conversely, lead to increased g_s under ample N supply, as has been shown in fertilization experiments (Li et al., 2004).

Intrinsic water-use efficiency (WUE_i) describes the relationship between g_s and the net photosynthetic C gain on the leaf level (Seibt et al., 2008). It can be derived from the analysis of foliar $\delta^{13}\text{C}$ signatures. In C_3 plants, $\delta^{13}\text{C}$ can be expected to vary between –12 ‰ if all stomates are

* Corresponding author.

E-mail address: markus.hauck@ecology.uni-freiburg.de (M. Hauck).

¹ 0000-0002-8218-8400

² 0000-0003-1545-2459

closed and g_s is minimal and -36‰ if carboxylation in the Calvin cycle obtains the dominant influence on C fractionation at fully opened stomates and maximum g_s (O'Leary, 1988). Strongly negative foliar $\delta^{13}\text{C}$ signatures $< -30\text{‰}$ (that correspond to low WUE_i) thus indicate little restrictions of CO_2 uptake by stomatal closure, which is usually the result of good water supply, but can be enhanced by high N availability (Shangguan et al., 2000), though effects of N on WUE_i have been found to be inconsistent (Brueck, 2008). Analyses of $\delta^{18}\text{O}$ signatures can additionally help to separate between effects of g_s or carboxylation on WUE_i (Sauer and Siegwolf, 2007), but were not part of the present study.

The influence of N on plant water relations can significantly modify the effect of climatic variability on growth. Even in strongly drought-limited, but also N-limited boreal forests in Central Asia in the ecotone between forest and steppe grasslands, the combined analysis of tree-ring width, foliar stable isotope ($\delta^{13}\text{C}$, $\delta^{15}\text{N}$) signatures and nutrient concentrations disclosed a tight control of annual stem increment by N availability in addition to water availability (Dulamsuren and Hauck, 2021).

The widespread use of mineral fertilizers, intensive livestock husbandry, fossil fuel combustion, and vegetation burning are significant sources of reactive N in the atmosphere that is deposited in forests and other ecosystems (Sommer et al., 2004; Ramanantenasoa et al., 2018; Liu et al., 2022). Anthropogenic N deposition has been discussed with respect to increases in forest productivity, for example, by increasing canopy leaf areas and foliar N concentrations (Du and de Vries, 2024), and accelerating litter decomposition (Berg and Matzner, 1997). In North America and Scandinavia, forests are even fertilized with N to stimulate wood formation (Hedwall et al., 2014). How the spatial variability of atmospheric N deposition interacts with tree water relations and thus with the drought tolerance of trees is much less investigated. Since the temperate forest zone of Europe is subjected to a considerable spatial variation in N deposition, we were interested in whether this variability has an effect on tree water relations and might therefore influence the susceptibility of Central European temperate forests to climate change-induced drought.

European beech (*Fagus sylvatica* L.), which would naturally dominate Central Europe's forest vegetation (Leuschner and Ellenberg, 2017), is a temperate tree species with a preference for cool-moist climate (Gessler et al., 2007) and high edaphic plasticity (Leuschner et al., 2006), which evolved in the Pleistocene (Denk, 1999; Jiang et al., 2022) and thrived during the Pleistocene's interglacial periods (Trájer, 2024) and since the early Holocene climate became moist enough to support beech forest (Tinner and Lotter, 2001). These specific climatic requirements, combined with the species' indifference to a wide range of edaphic conditions, makes European beech to an obvious candidate of a species that is susceptible to climate change, but little vulnerable to other changes in site conditions. Correspondingly, evidence is accumulating that wood formation in European beech is predominantly limited by low water availability and that steady increases in VPD due to the constant rise of temperatures since the 1970s and the increased occurrence of hot summer droughts reduces the productivity of European beech forests (Martínez del Castillo et al., 2022; Klesse et al., 2024; Leifsson et al., 2024) and is the cause of increased tree mortality (Schuldt et al., 2020; Dulamsuren et al., 2022). Survival and growth of *F. sylvatica* seedlings and saplings are also reduced at high air temperature and low air humidity (Lendzion and Leuschner, 2008; Muffler et al., 2021; Zweifel et al., 2021).

Nonetheless, there is also indication of negative effects of high N supply on European beech. In mature *F. sylvatica*, Hess et al. (2018) found a positive effect of precipitation on annual radial stem increment in consistence with many other tree-ring studies (e.g. Scharnweber et al., 2011; Zimmermann et al., 2015; Dulamsuren et al., 2017), but a negative effect of high temperatures only in combination with high N deposition. Two years of NH_4NO_3 fertilization with $50\text{ kg N ha}^{-1}\text{ a}^{-1}$ in 5 year-old saplings of ca. 80 cm height significantly reduced height increment (Dziedek et al., 2016). Dietrich et al. (2024), (2025)

demonstrated decreasing shoot length increment and reduced root/-shoot ratios in *F. sylvatica* seedlings and saplings in 54 forests of Germany with increasing atmospheric N deposition along a gradient of $23\text{ kg N ha}^{-1}\text{ a}^{-1}$ ($9 - 32\text{ kg N ha}^{-1}\text{ a}^{-1}$). By contrast, precipitation, temperature or meteorological drought intensity (Standardized Precipitation-Evapotranspiration Index, SPEI) exerted no significant effect. Reduced growth of beech regeneration at high N loads (instead of a positive fertilization effect) could be explained by potential reduction of the root system (Ågren and Franklin, 2003) and decreasing ectomycorrhization at high N supply (de Witte et al., 2017), which both would limit water uptake under dry conditions. Furthermore, high N loads can increase competition by the ground vegetation and reduce light availability on the forest ground due to a denser canopy (Dirmböck et al., 2014).

Since the results of Dziedek et al. (2016), Hess et al. (2018), and Dietrich et al. (2024) suggested that European beech is sensitive to high N loads in a range that occurs in regions with intense agriculture in Europe (Braun et al., 2022), we were interested to what extent WUE_i of *F. sylvatica* is influenced by atmospheric N deposition. A negative influence of high N deposition on WUE_i would imply an extension of periods with open stomates and CO_2 assimilation and can thus contribute to increased biomass production (Dulamsuren and Hauck, 2021; Huang et al., 2021), but increases the risk of temporary or permanent drought damage. We, therefore, established a network of 45 *F. sylvatica* stands across Germany in the core area of the distribution range of European beech with the intention to cover wide ranges of atmospheric N deposition, mean annual precipitation (MAP), and mean annual temperature (MAT) and assessed WUE_i in beech regeneration with the help of foliar $\delta^{13}\text{C}$ signatures (Farquhar et al., 1982). Because foliar $\delta^{13}\text{C}$ in mature *F. sylvatica* has been shown to increase (suggesting increasing WUE_i) with decreasing MAP along a gradient of 450 mm (ranging from 520 – 970 mm) in northern and central Germany (Meier and Leuschner, 2008), we hypothesized (1) that foliar WUE_i and $\delta^{13}\text{C}$ signatures along our wider climatic gradients would be primarily influenced by MAP, but that a secondary effect of N deposition on WUE_i and $\delta^{13}\text{C}$ would be detectable. Furthermore, we hypothesized (2) that both high MAP and high N deposition would be associated with low $\delta^{13}\text{C}$ signatures and low WUE_i . In addition, we analyzed the relationship between the foliar $\delta^{13}\text{C}$ signature and the leaf K content, assuming (in addition to non-structural carbohydrates and amino acids) a significant role of K^+ in the osmoregulation (Szczerba et al., 2009; Argiolas et al., 2016) of beech, which is an anisohydric species (Leuschner et al., 2022). We therefore tested the hypothesis (3) that low $\delta^{13}\text{C}$ signatures and low WUE_i are associated with high K concentrations in the leaves of *F. sylvatica*.

2. Material and methods

2.1. Study design

The study was carried out in 45 stands of European beech (*Fagus sylvatica* L.) in Germany, which were selected according to their atmospheric N deposition, MAP, and MAT (Fig. 1). The stands covered a range of $21\text{ kg N ha}^{-1}\text{ a}^{-1}$ ($10.5 - 31.4\text{ kg N ha}^{-1}\text{ a}^{-1}$) for N deposition, 1130 mm (560 – 1690 mm) for MAP, and 3.4°C ($6.8 - 10.2^\circ\text{C}$) for MAT (Table S1). Within each beech stand, we selected 3 circular plots with a radius of 10 m and a minimum distance of 40 m between the plots. On each plot, 3 saplings with a maximum height of 50 cm and an age of 5 – 10 years were selected as sample trees. The forests were all managed as high forests and the studied beech saplings were from natural regeneration from seeds throughout.

2.2. Sample collection, stable isotope and element analysis

A total of 405 beech saplings was collected in the summer of 2021. The complete foliage of each sapling was dried at 70°C for at least 72 h and homogenized in a swing mill. Stable isotope analysis for $\delta^{13}\text{C}$ were

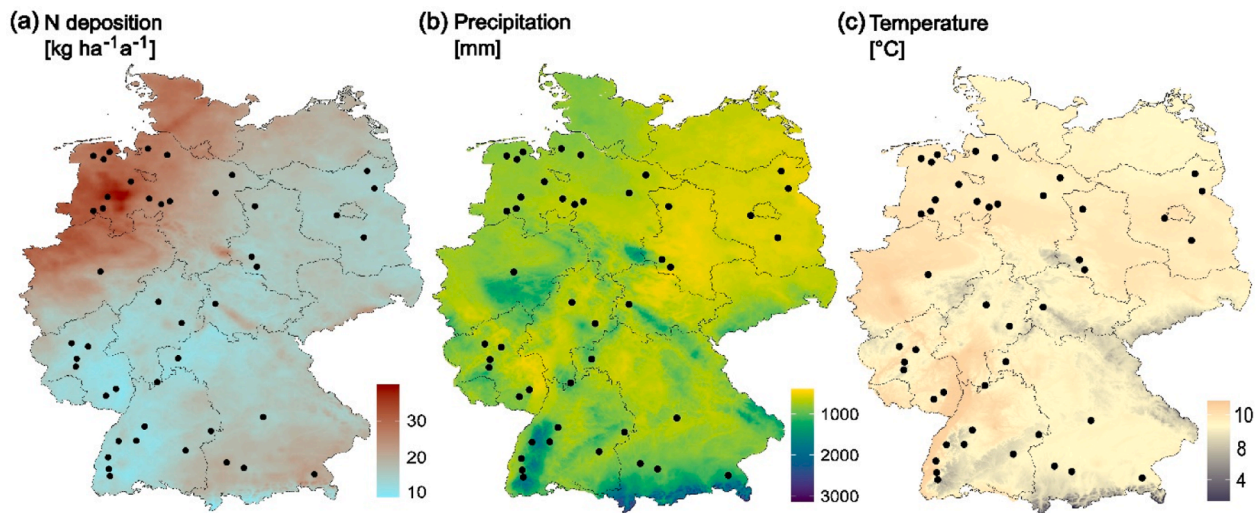


Fig. 1. Spatial variation of (a) atmospheric N deposition, (b) mean annual precipitation (MAP), and (c) mean annual temperature (MAT) in Germany and location of studied forest stands.

conducted in the Center for Stable Isotope Research and Analysis, University of Göttingen, Germany. These analyses were done using an isotope ratio mass spectrometer (Delta V Advantage, Thermo Electron, Bremen, Germany), a ConFlo III interface (Thermo Electron, Bremen, Germany) and an element analyzer (Flash 2000, Thermo Fisher Scientific, Cambridge, UK). High (i.e. less negative) $\delta^{13}\text{C}$ signatures indicate stomatal closure (i.e. low g_s), because the $^{13}\text{CO}_2$ is less discriminated in the Calvin cycle under low leaf-internal CO_2 concentration (Farquhar et al., 1982). The concentration of K (as an important osmolyte; Argiolas et al., 2016) was determined after acid digestion of ground leaves with H_2SO_4 at 190°C for 8 h under pressure. Chemical analysis was conducted using inductively coupled plasma optical emission spectroscopy (ICP-OES, Ultima 2, Horiba, Longjumeau, France).

2.3. Calculation of intrinsic water-use efficiency (WUE_i)

In a first step, foliar $\delta^{13}\text{C}$ signatures were used to calculate the isotopic discrimination ($\Delta^{13}\text{C}$, in ‰) between the CO_2 in the atmosphere and the plant following the equation: $\Delta^{13}\text{C} = (\delta^{13}\text{C}_{\text{atmosphere}} - \delta^{13}\text{C}_{\text{plant}}) / ((1 + \delta^{13}\text{C}_{\text{plant}}) / 1000)$ (Farquhar and Richards, 1984). Since field sampling took place in 2021, $\delta^{13}\text{C}_{\text{atmosphere}}$ was estimated as -8.7‰ based on Graven et al. (2020). In a second step, WUE_i (in $\mu\text{mol CO}_2 \text{ mmol}^{-1} \text{ H}_2\text{O}$) was calculated from $\Delta^{13}\text{C}$ using the equation $\text{WUE}_i = (C_a (b - \Delta^{13}\text{C})) / (1.6 (b - a))$. In this equation, C_a is the atmospheric concentration of CO_2 in 2021 of 416.41 ppm (https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_annmean_mlo.txt), a is the ^{13}C fractionation during CO_2 diffusion through the stomates (4.4 ‰), and b is the ^{13}C fractionation by Rubisco (ribulose-1,5-bisphosphate carboxylase/oxygenase) and PEP (phosphoenolpyruvate) carboxylase (27 ‰) (Farquhar and Richards, 1984; O'Leary, 1988).

2.4. Nitrogen deposition and climate data

Modeled data of atmospheric N deposition (PINETI-3; German Environment Agency, Dessau-Roßlau, Germany), MAP, and MAT (German Meteorological Service, Climate Data Center, Offenbach, Germany) were available at a resolution of $1 \text{ km} \times 1 \text{ km}$. N deposition data were modeled for broadleaved forest and covered the period 2013 – 2015. Climate data refer to 1991 – 2020. Most of the variation of N deposition in our dataset derived from NH_4^+ deposition (Dietrich et al., 2024), which primarily originates from the agricultural sector (Aksoyoglu et al., 2020). The data on N deposition is the latest available for Germany. There is a follow-up project (PINETI-4) to extend the

dataset from 2015 to 2019 (Kranenburg et al., 2024), but this dataset is not yet publicly available. Mean N deposition in Germany showed some interannual variability between 14 and $16 \text{ kg N ha}^{-1} \text{ a}^{-1}$, but no consistent trend for increase or decrease (Kranenburg et al., 2024).

2.5. Statistical analysis

Data were analyzed with R 4.4.1 software. Arithmetic means $\pm$ standard errors (SE) are given throughout the paper. Tree-related data were pooled in two steps for each stand (first calculating plot-related means and then stand-related means), if the independent parameters for the analysis were only available on the stand level. The effect of N deposition, MAP, and MAT and their interaction was analyzed with linear mixed effect modeling using the R package 'lme4' 1.1–35–5. N deposition was square-root transformed to remove a right-skewed data distribution. Independent variables were rescaled before analysis using the 'scale' function in R and tested for multicollinearity with the R package 'performance' 0.12.4. The interaction of N deposition and MAT was excluded from the model due to variance inflation factor (VIF) of 4.8 indicating multicollinearity. The remaining predictors in the model had VIF values between 1.2 and 2.4 and thus even fulfilled the strict criteria of Zuur et al. (2010) for avoiding multicollinearity. Removing more predictors did not decrease the Akaike Information Criterion (AIC). Normal distribution of the residuals was checked positively with the R package 'DHARMa' 0.4.6 using quantile-quantile plots.

3. Results

3.1. Influence of N deposition, precipitation, and temperature on foliar $\delta^{13}\text{C}$ and WUE_i

The $\delta^{13}\text{C}$ signatures and WUE_i of *F. sylvatica* leaves decreased with increasing atmospheric N deposition (Fig. 2a, b). There was no relationship between foliar $\delta^{13}\text{C}$ or WUE_i with MAP (Fig. 2c, d) and MAT (Fig. 2e, f). Linear mixed effect modeling confirmed this result (Table 1). Atmospheric N deposition had a highly significant ($P \leq 0.001$) negative effect on WUE_i . By contrast, MAP ($P = 0.96$) and MAT ($P = 0.82$) were not related to WUE_i . In addition to the correlations in Fig. 2, the mixed effect model also showed that there was no interaction between N deposition, MAP, and MAT that exerted a significant effect on WUE_i .

Overall foliar $\delta^{13}\text{C}$ signatures were quite low and amounted to $-33.45 \pm 0.07\text{‰}$ across all study sites. Mean $\Delta^{13}\text{C}$ was $25.61 \pm 0.07\text{‰}$ and mean WUE_i was $23.96 \pm 1.26 \mu\text{mol mol}^{-1}$.

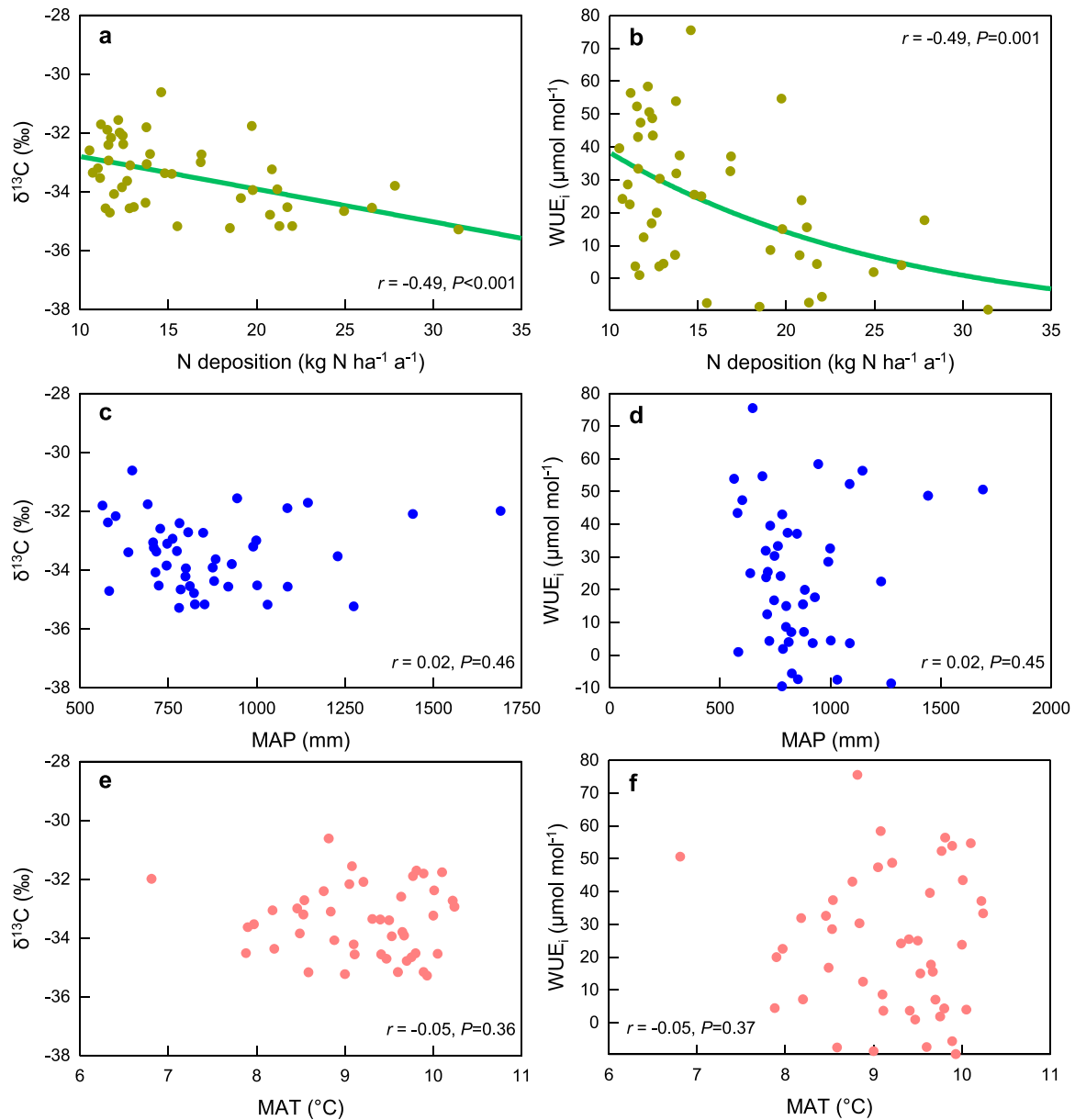


Fig. 2. (a, c, e) Foliar $\delta^{13}\text{C}$ signatures and (b, e, f) intrinsic water-use efficiency (WUE_i) in *Fagus sylvatica* in variation of (a, b) atmospheric N deposition, (c, d) mean annual precipitation (MAP), and (e, f) mean annual temperature (MAT) in 45 forest stands across Germany.

3.2. Relationship between foliar $\delta^{13}\text{C}$ and K content

The $\delta^{13}\text{C}$ signatures decreased with increasing K content of the beech leaves (Fig. 3). There was no relation between the foliar K content and atmospheric N deposition, MAP, or MAT (Fig. S1).

4. Discussion

The between-stand variation of WUE_i and the foliar $\delta^{13}\text{C}$ signatures in European beech regeneration growing in 45 forest stands across Germany in the center of the distribution range of this important Central European tree species was significantly related to atmospheric N deposition, but not to the long-term means of precipitation and air temperature, which does not confirm our first hypothesis assuming a dominant influence of the precipitation gradient of WUE_i . At first glance, this dominant effect of N deposition challenges the image of a tree species imminently driven to its physiological limits by drought and heat as the result of climate change, which is prevalent in the present literature

(Gillner et al., 2013; Leuschner et al., 2023; Leifsson et al., 2024). Although it is undisputed that stemwood production in European beech is increasingly limited by dry and warm conditions (Martinez del Castillo et al., 2022; Klesse et al., 2024), this limitation is apparently less valid for WUE_i in the regeneration of beech growing under the shady canopy of beech forests.

The different response of foliar $\delta^{13}\text{C}$ signatures and WUE_i to MAP compared to mature European beech (Meier and Leuschner, 2008) can be attributed both to differences in plant water relations and in the carboxylation efficiency between the sun crown of mature trees and the shady forest interior below the canopy. Firstly, the regeneration in the understory is exposed to lower VPD values than the sun crown of mature trees and can thus be assumed to maintain higher g_s (Lendzion and Leuschner, 2008). Secondly, small tree individuals in the understory are exposed to higher atmospheric CO_2 concentrations from plant and soil respiration than mature trees in the upper canopy, which leads to a stronger depletion in ^{13}C in the plant tissues (Francey and Farquhar, 1982; Ekblad et al., 2005; Peñuelas et al., 2008).

Table 1

Linear mixed model for the effects of atmospheric N deposition, MAP, and MAT on intrinsic water-use efficiency (WUE_i) in *Fagus sylvatica*.

Parameter	Estimate	SE	<i>t</i>	<i>P</i>	CI	
					2.5 %	97.5 %
Fixed effects						
(Intercept)	22.27	2.92	7.64		16.79	27.74
N deposition ^a	-12.79	3.07	-4.16	< 0.001	-18.57	-7.04
MAP	-5.36	4.21	-1.27	0.96	-13.26	2.55
MAT	3.02	2.87	1.05	0.82	-8.43	9.14
N deposition ^a × MAP	-9.34	5.22	-1.79	0.55	-19.13	0.47
MAP × MAT	-0.87	1.65	-0.52	0.39	-4.01	2.23
Random effects						
Stand ^b	17.8					
Residual ^b	14.5					
Conditional R ²	0.69					
Marginal R ²	0.21					

^a Square-root transformed due to right-skewed distribution.

^b Estimates of random effects represent standard deviations (SD).

The mean $\delta^{13}\text{C}$ signature of -33.45 ± 0.07 ‰ is rather close to the value of -36 ‰ that is to be expected at maximal g_s when only carboxylation is limiting WUE_i (O'Leary, 1988). The origin of the sample from the regeneration thus explains why foliar $\delta^{13}\text{C}$ signatures and WUE_i were neither related to MAP nor to MAT, although European beech is known as a species that is under urgent pressure due to climate change-induced drought and heat based on data from mature trees (Leuschner et al., 2023; Hauck et al., 2025). However, these results do not imply that also beech regeneration cannot be driven to even lethal limits due compound drought and heat (Bolte et al., 2016). A significant effect of the summer SPEI in 2018 on the beech regeneration density in 2021 in our study plots (Dietrich et al., 2024), suggested the occurrence of a mortality burst in beech seedlings and saplings due to the extreme hot drought in Central Europe in 2018.

The negative effect of N deposition on foliar $\delta^{13}\text{C}$ and WUE_i, which is accordance with our second hypothesis, suggests that high N availability even reinforces the already low limitation of WUE_i due to stomatal resistance in the beech regeneration. N deposition seems thus to be a factor that is of high importance for the water and carbon balance of beech, the significance of which has been underestimated. Previous studies pointing to negative effects of high N availability on European beech in young (Dziedek et al., 2016; Dietrich et al., 2024) and mature (Hess et al., 2018) trees primarily assumed effects of biomass allocation to above- and belowground organs. The decrease of foliar $\delta^{13}\text{C}$ signatures and WUE_i with increasing N deposition found in our study, however, suggests that high N deposition causes the beech regeneration to keep their stomates open for longer periods than under similar moisture conditions without anthropogenic N loads. Comparative measurements of foliar $\delta^{13}\text{C}$ signatures along N deposition gradients lack for mature trees of European beech. Such analyses would most likely yield a higher influence of climate on WUE_i than in the regeneration (Meier and Leuschner, 2008), but could elucidate to what extent N deposition is also interfering with the drought tolerance of mature beech. Reductions of shoot length growth and the root/shoot ratio with increasing N deposition (Dietrich et al., 2024) as well as significantly reduced fine-root tip frequencies and specific root lengths with increasing N deposition at low MAP (Dietrich et al., 2025) in the beech regeneration on our sample plots across Germany suggest that high N deposition affects through several mechanism the vitality of young European beech trees under drought.

Reduced stomatal resistances, i.e. increases in g_s , with increasing N availability were mostly demonstrated under laboratory and greenhouse conditions (Ghashghaie and Saugier, 1989; Huang et al., 2021), but recently they have also been substantiated (based on $\delta^{13}\text{C}$ signatures) in boreal forest stands of *Larix sibirica* Ledeb. with varying N deposition (Dulamsuren and Hauck, 2021). Increases in g_s that are not triggered by

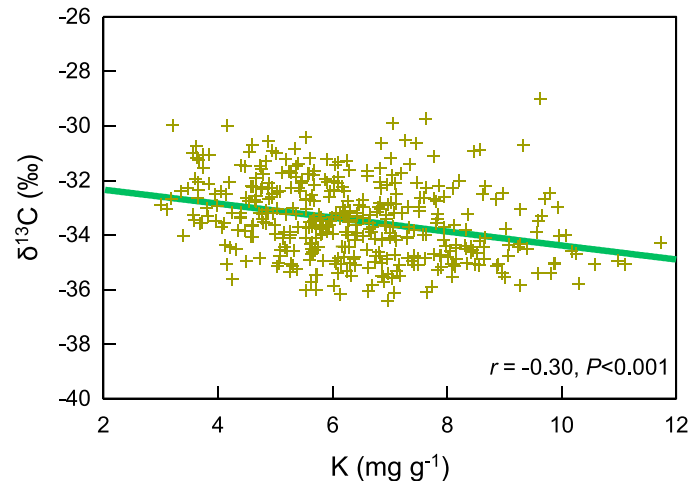


Fig. 3. Relationship between the foliar K content and $\delta^{13}\text{C}$ signatures in *Fagus sylvatica* ($N = 405$ trees).

VPD or soil moisture, but by N availability can have positive effects on net photosynthesis and thus growth, as has been demonstrated in *L. sibirica* with respect to annual stem increment, but at the cost of lowered WUE_i. If WUE_i is reduced and stomatal opening is stimulated by N, this poses a risk during drought, if N-induced high g_s causes greater water losses in sensitive tissues and in the xylem due to high transpiration than would be expected if the stomata were closed earlier without the control by N. This relationship between N availability and tree water relations suggests that N deposition should be integrated into the discussion on the climate change sensitivity of tree species and the climate change adaptation of forests to a much greater extent than so far.

Our results refer to the effects of the spatial variation of N deposition and climate on WUE_i. Of course, VPD and soil moisture remain the most important influencing factors for g_s , depending on the daily course of evapotranspiration and on weather conditions (Köcher et al., 2009), but also on hydrological soil conditions (Weigel et al., 2023). The decrease of $\delta^{13}\text{C}$ signatures (indicating stomatal opening) with increasing foliar K content, which is consistent with our third hypothesis, points to active osmotic adjustment by K^+ (and other substances that were not analyzed in our study) in response to increased VPD and/or reduced soil moisture (Leuschner, 2020; Weithmann et al., 2022). However, our results show that narrowing the study of water and carbon relations in European beech (and other tree species) to the influence of atmospheric humidity and soil moisture does not give the complete picture. The need to account for atmospheric N deposition to assess the water and carbon relations in European beech adds complexity to attempts to predict the fitness of this important tree species in the age of climate change. There is no doubt that beech is sensitive to drought and heat and thus to climate change (Walthert et al., 2021; Hauck et al., 2025), yet there is considerable uncertainty in predicting its likelihood of survival and its growth performance in a warmer and drier climate in regions with high N deposition, which calls for the need of an extension of studies on WUE_i–N availability relationships from saplings to mature beech trees.

5. Conclusions

Our results of WUE_i from beech regeneration indicate together with studies of biomass allocation and root morphology (Dietrich et al., 2024, 2025) that European beech is more sensitive to drought at high N deposition than under N-poor conditions. Since intensive livestock husbandry is a key source of high N deposition in Central Europe (Aksoyoglu et al., 2020) and a fundamental reduction of N emissions from the agricultural sector is not in sight, attempts for the climate change adaptation of forestry should include more than so far interactions between N deposition and the drought- and heat-tolerance of

trees. However, it is too early to give detailed recommendations for forest management at the present state of knowledge, as the drought and heat sensitivity of *F. sylvatica* itself needs more exploration and because there is a lack of data of drought-N deposition interactions in mature beech trees.

Author contributions

Both authors developed the concept, analyzed data and wrote the paper. VD carried out field work.

CRedit authorship contribution statement

Markus Hauck: Writing – review & editing, Writing – original draft, Visualization, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Viktoria Dietrich:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization.

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Declaration of Competing Interest

We declare no conflict of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.foreco.2025.122835](https://doi.org/10.1016/j.foreco.2025.122835).

Data availability

Data will be made available on request.

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